

Energy Based Constraint Analysis finds wing loading $\left(\frac{W}{S}\right)$ & T to W ratio $\left(\frac{T}{W}\right)$ for the entire aircraft

• Stability / Trim

tail volume constraint, $S_H = \frac{V_H S_w \bar{c}}{l_H}$

- $\bar{c} = \frac{S_{wing area}}{b}$

- V_H = horizontal tail volume coefficient, choose by aircraft type

- l_H = horizontal tail moment arm

• Can graph the tail ~~design constraint plot~~ ~~static margin study~~

- x-axis tail arm y-axis required tail area

then choose

- Constraints max fuselage length, minimum static margin, max tail weight, elevator authority requirement

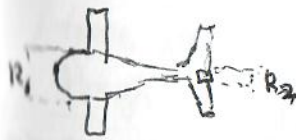
- static margin tells how far the CG is ahead of neutral point

- static margin vs tail volume

- tail weight vs tail arm

This page helps understand pg. 6

2. Tail moment arm, $l_H = \sqrt{\frac{2 S_{ref} (V_H C_{ref} + V_H l_{ref})}{\pi (R_1 + R_2)}}$ For tail moment arm & empennage



- V_H : horizontal stabilizer tail volume coefficient how effective vs it is at controlling pitch & providing longitudinal stability

- C_{ref} : MAC

- b_{ref} : wing span

- R_1 : fuselage radius

- R_2 : fuselage radius near tail

2 Tail horizontal area $S_H = \frac{V_H S_w \bar{c}}{l_H}$

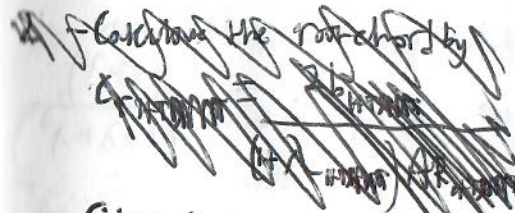
3 Vertical stabilizer area $S_v = \frac{V_v S_w b_{wing ref}}{l_v}$

4 Choose AR for vertical & horizontal stabilizer & their λ (taper ratio)

1 - Calculate wing span for both stabilizer by

$$b_{stabilizer} = \sqrt{AR_{stabilizer} S_{stabilizer}}$$

* Calculation of c_r was for trapezoidal wing, we are designing a tapered wing



$c_t = \lambda c_r$ can also be done for horizontal/vertical stabilizer to wing for root tip chord to span